



From AI to Agentic AI

The Role of Open Standards

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A Tale of Two Internets

The web scaled on three foundations. The Internet of Agents needs the same three — one layer up.

INTERNET OF INFORMATION

The Web · 1990s

TCP/IP, HTTP, HTML — a common language any computer could speak, owned by no vendor.

Stateless HTTP and DNS let any server join and scale — no central coordinator, no permission required.

IETF, W3C, ICANN — standards stewarded in the open, controlled by nobody.



INTERNET OF AGENTS

Agentic AI · 2020s

MCP, A2A, agentic commerce protocols — one open interface per layer of the agentic stack.

Stateless agent-tool connections, registries, and gateways — agents scale across vendors and clouds without permission.

AAIF at the Linux Foundation — a neutral home where the agentic stack is governed by no single company.

**OPEN
PROTOCOLS**

**STATELESS
INFRASTRUCTURE**

**NEUTRAL
GOVERNANCE**

Every internet needs a protocol anyone can implement, an architecture that scales without permission, and governance nobody owns.

The Evolution of AI

Three phases brought us here. A fourth is beginning now.

1

BEFORE 1990

Rule Based Deterministic Systems

Expert systems encoded human knowledge as explicit if then rules. Every behavior was written by hand.

THE LIMIT

Brittle. Anything outside the rules broke the system.

2

1990s TO 2016

Statistical Machine Learning

Neural networks and Boltzmann machines learned patterns from data instead of handwritten rules.

THE LIMIT

Needed large labeled datasets, and each model solved only one narrow task.

3

2017 TO TODAY

Transformers and LLMs

Training became largely unsupervised and at scale, producing general models that transfer across tasks.

THE UNLOCK

Reasoning, chat, and general purpose assistants.

4

NOW EMERGING

Agentic Systems

Systems that plan, use tools, observe, take action and iterate. They pursue goals across many steps instead of only answering questions.

The Inflection Point

LLMs didn't just make AI faster. They gave it the ability to reason — and reasoning made agency possible.

Reasoning at scale

LLMs can generalize across novel situations without being explicitly programmed — deciding what to do next in contexts they've never seen.

Tool use

Models can call APIs, search the web, write and execute code, and interact with external systems — not just generate text.

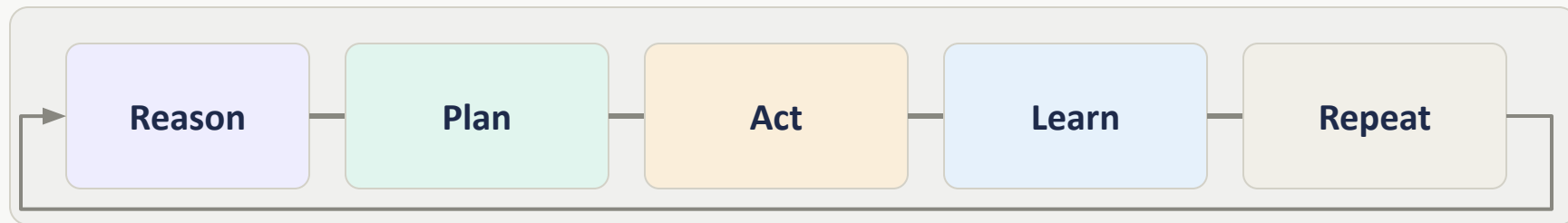
Memory & context

Longer context windows and external memory let agents maintain state across long tasks — remembering what they did and learned.

Every previous AI system needed a human to define the steps. LLMs were the first that could figure out the steps themselves.

Agentic AI defined

The loop is what's new. Agents don't just respond — they reason, plan, act, observe, and repeat. Autonomously. At scale. With consequences.



🔄 autonomous loop — no human required at each step

Consequence-bearing

Agents book flights, execute code, send emails, move money. Errors are real. Stakes are real.

Long-horizon

Tasks run for minutes or hours — far beyond a single prompt-response exchange.

Multi-agent

Agents spawn subagents, delegate, coordinate — a network of autonomous systems working together.

Why Open Standards are the Only Solution?

Systemic Risk

- Agentic AI touches every business — finance, health, logistics, government.
- It will run critical decisions, transactions, and infrastructure at global scale.
- Concentrated ownership by a few private actors = control of critical infrastructure.
- We do not let one company own the power grid. The same logic applies here.

Interoperability

- Agents need to interoperate across vendors, stacks, and industries.
- The value is network value — it lives in the connections, not the nodes.
- Proprietary walled gardens cannot compound. They fragment and stall.
- Open protocols (MCP, A2A, ACP) collapse N×M integrations into one interface.

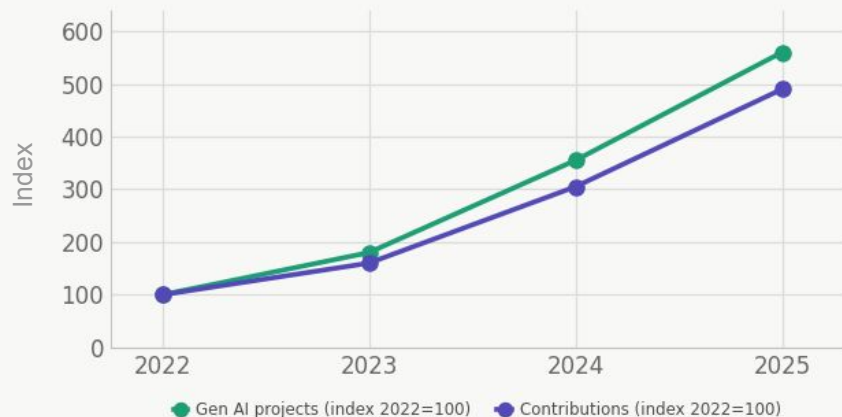
Trust

- Open source code gets eyes from a global community — many reviewers, fewer blind spots.
- Vulnerabilities are found and patched faster than in closed systems.
- Features and operational hardening evolve in the open, driven by real production use.
- Lower risk of hidden defects, supply-chain compromise, or security incidents.

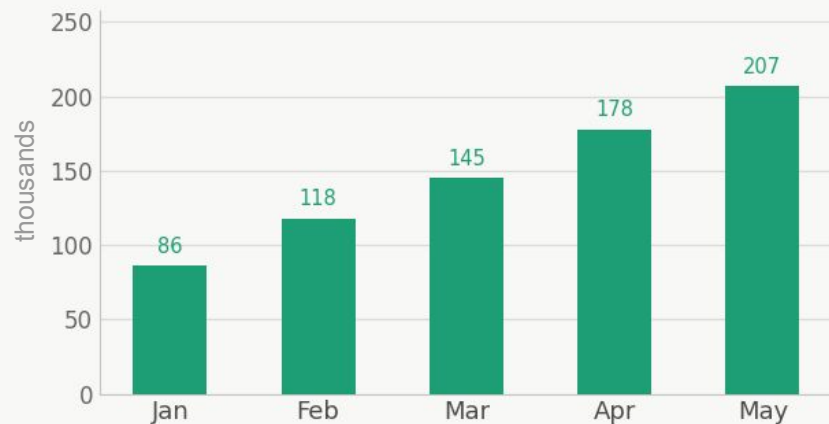
Open source and open standards are the only foundation that scales — safely, together, at network value.

Developers are voting with their commits

Open source AI contribution growth on GitHub, 2022–2025. The community is building the future in the open.



Indexed growth (2022 = 100)



Monthly gen AI contributors, 2025 · May peak 206,830 (+132% YoY)

+178%

Public repos importing an LLM SDK (YoY)

89%

Of orgs using AI use open source (LF Research 2025)

4.3M+

AI repositories now on GitHub

The Agentic Stack

L4 is AAIF's primary scope • L1–L3 are coordinated with LF AI, CNCF, PyTorch Foundation • Ecosystem partnership

Layer 5: User Application

Chat & voice UIs, IDE plugins, embedded experiences, agent storefronts

Layer 4: Agentic Platform

Agentic development and runtime, protocols, data & knowledge, governance & control, observability & reliability, Trust & Safety.

Layer 3: Intelligence

Foundation models, fine-tuning, inference, embeddings, model serving, quantization

Layer 2: Software Infrastructure

Kubernetes, containers, service mesh, message queues, databases

Layer 1: Hardware Infrastructure

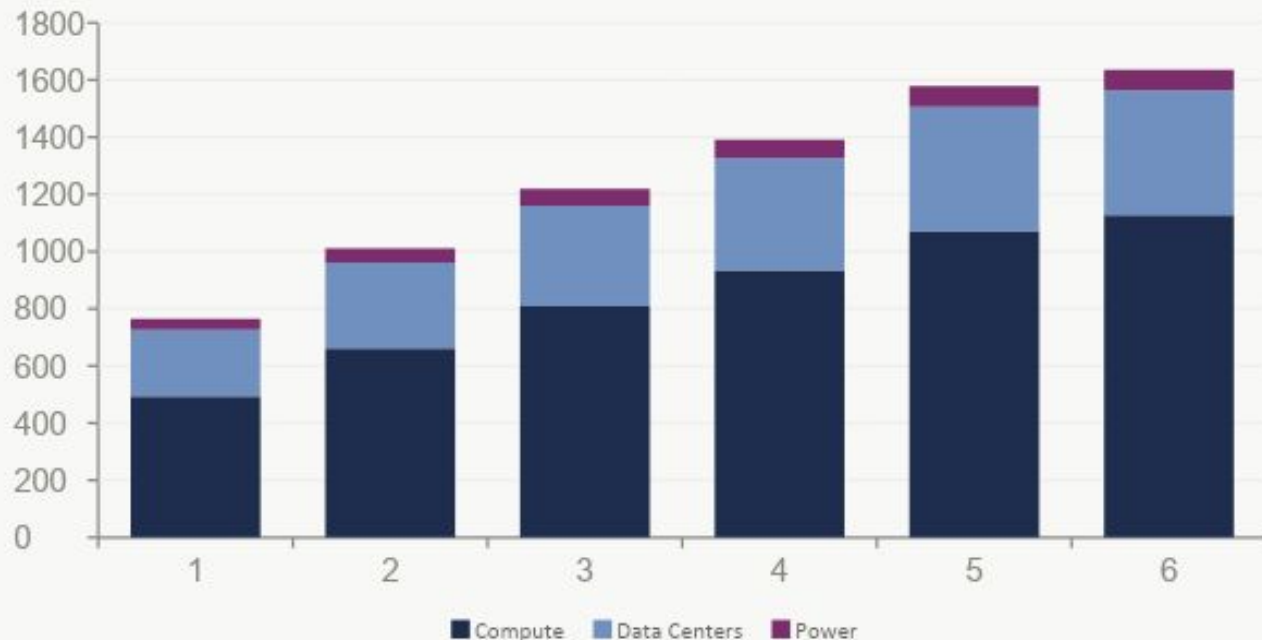
GPUs, TPUs, custom accelerators, networking, edge devices



Agentic AI
Foundation

Layer 1: AI infrastructure: a \$7.6T capital commitment, 2026–2031

Goldman Sachs / NVIDIA projections. Baseline aggregate AI CapEx (\$B) across compute, data centers, and power. X-axis: 1 = 2026 → 6 = 2031.



~\$7.6T

Total CapEx 2026–2031

\$765B → \$1.6T

Annual growth

~65%

Compute share

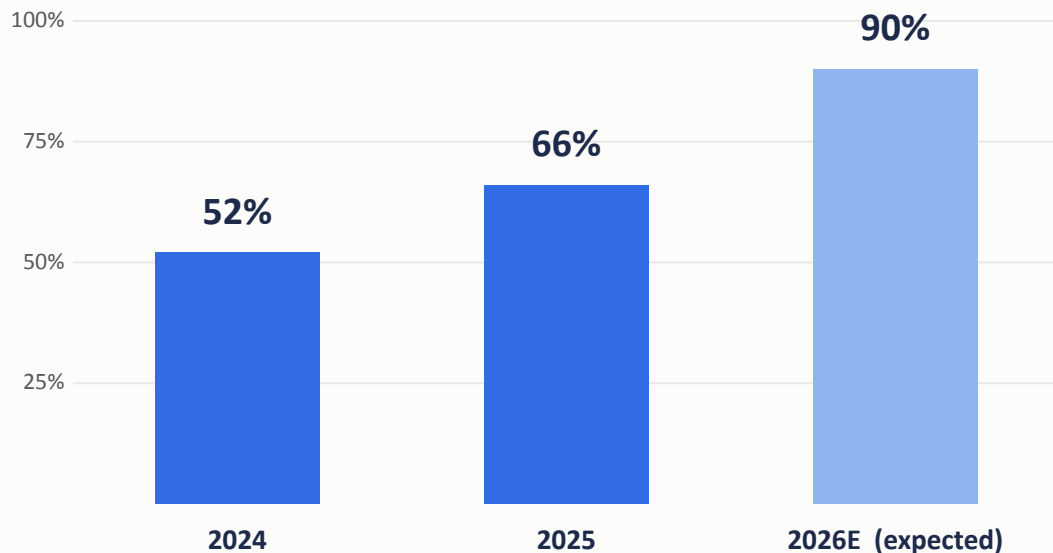
\$765B

2026 spend alone

Layer 2: Kubernetes is the operating system for AI

GenAI inference workloads running on Kubernetes — CNCF Annual Cloud Native Survey

GenAI orgs running inference on Kubernetes



82%

production Kubernetes adoption in 2025, up
from 66% in 2023

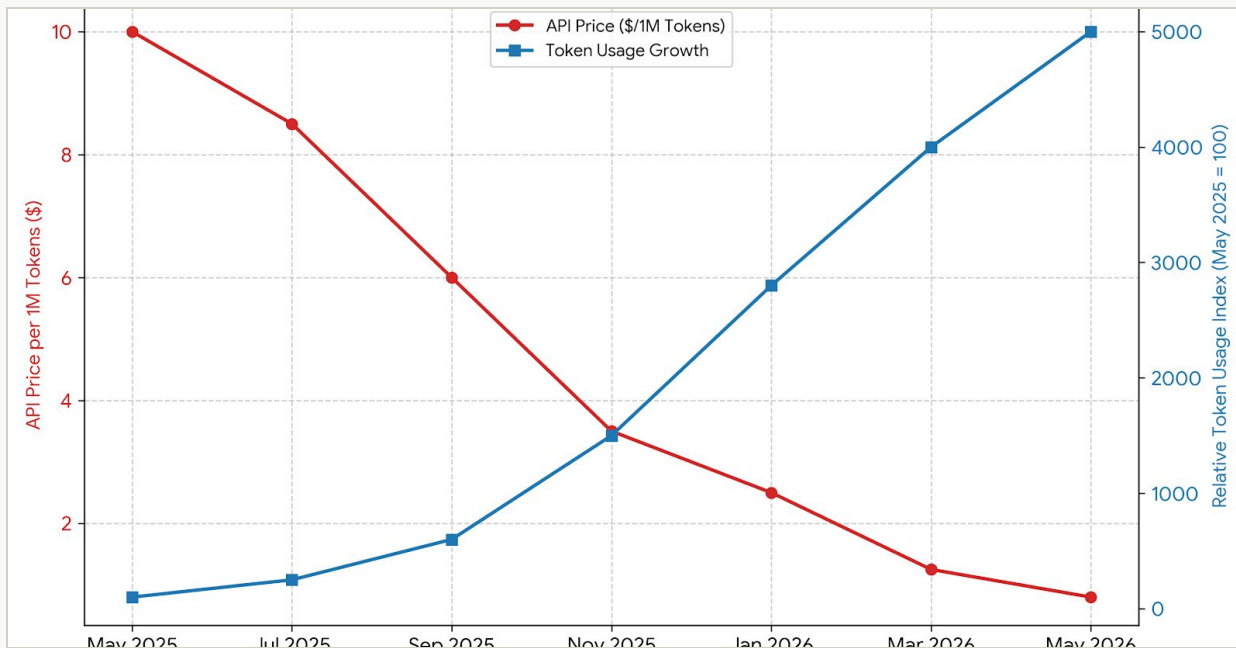
\$4.8B → \$22.6B

Kubernetes-for-AI market, 2025–2034
(18.8% CAGR)

CNCF positions Kubernetes as the operating system for AI. If MCP is the protocol, Kubernetes is the runtime.

Layer 3: AI Token Usage vs API Price Drops

As efficiency increases and prices fall, demand for tokens explodes, illustrating the Jevons Paradox in AI development.



Price Drop

-90%

API Price per 1M Tokens

Usage Surge

50x

Relative Token Usage Growth

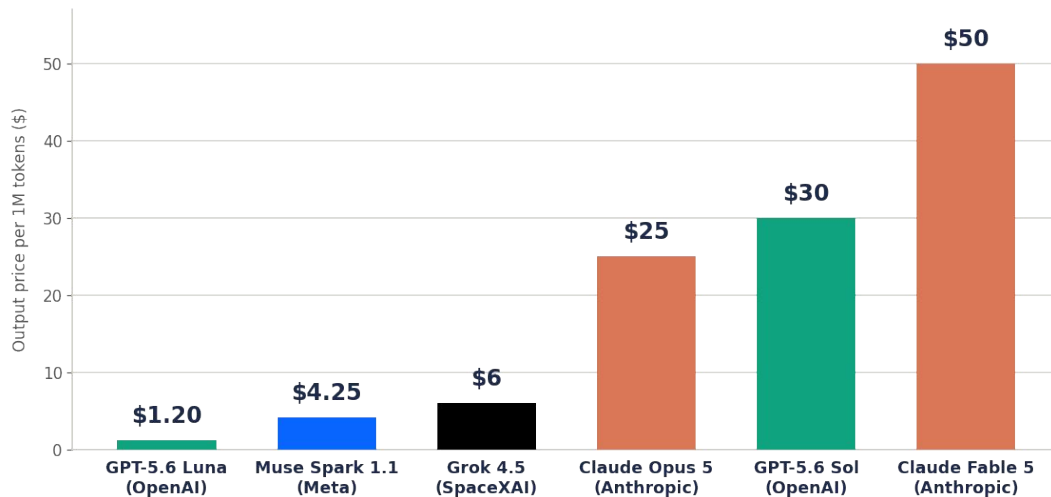
KEY TAKEAWAY

Prices have fallen for three straight years while usage keeps accelerating. Cheaper tokens are not shrinking spend — they're unlocking far more consumption

Pick the Best Model or Use a Model Portfolio

The race to the bottom keeps going — OpenAI cut GPT-5.6 Luna to \$1.20/M output on Jul 30, 2026, now the cheapest frontier-lab model on the board.

Frontier model API pricing — August 2026



Different Strategies

OpenAI

Tiered price-performance segmentation

SpaceXAI (xAI)

Speed & efficiency at flash-model pricing

Meta

Aggressive entry by attacking margins

Anthropic

The opposite bet: premium for quality

Layer 3: Open versus Closed Models

DEC 2026

From extrapolated trendline: open closes the gap with closed

Now within 3 points: Kimi K3 (60) vs Claude Opus 5 (63), Aug 2026

FRONTIER OPEN MODELS, 2026

- Kimi K3 (2.8T, 104B active)
- Qwen3.8 Max (2.4T, 95B active)
- DeepSeek V4 Pro (1.6T, 49B active)
- GLM-5.3 (743B, weights Aug 28)

LEADING US OPEN MODEL

- Inkling 975B, Apache 2.0 (AAII 42)

THE OPEN-MODEL ADVANTAGE

6×

what closed models cost vs.
comparable open (MIT/LF)

80%

of model usage still runs on
closed models (OpenRouter)

\$24.8B

in unrealized value/yr; range
\$20–48B (MIT/LF)

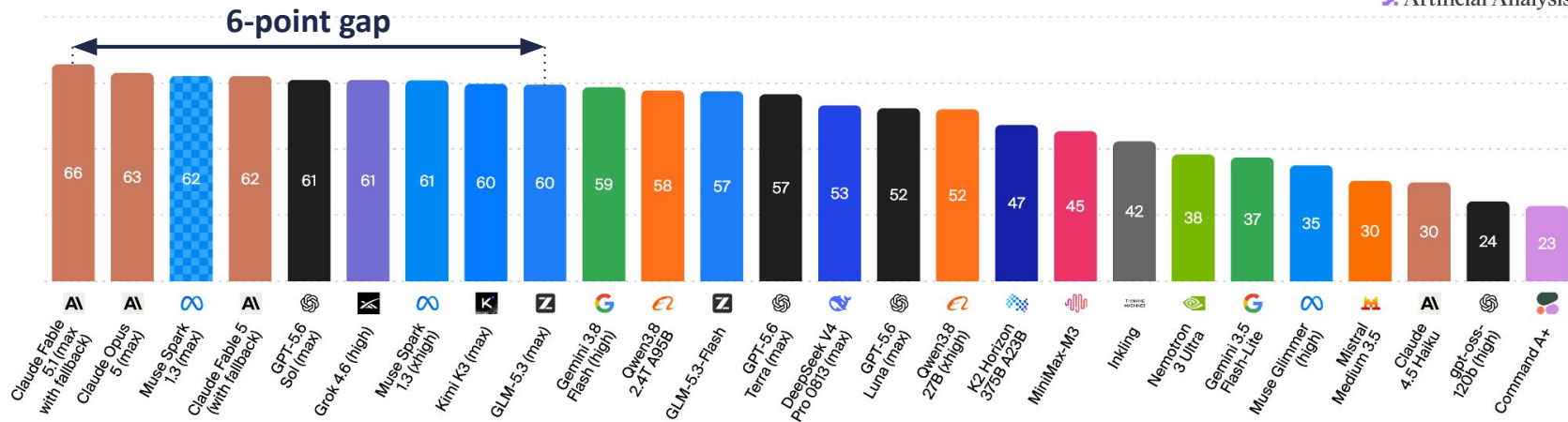
Open Models Are Closing the Gap

Artificial Analysis Intelligence Index [↗](#)

Artificial Analysis Intelligence Index v4.1.1 incorporates 9 evaluations: GDPval-AA v2, τ^3 -Banking, Terminal-Bench v2.1, SciCode, Humanity's Last Exam, GPQA Diamond, CritPt, AA-Omniscience, AA-LCR

Not publicly available

Artificial Analysis



Source: Artificial Analysis Intelligence Index v4.1.1, snapshot 17 Sep 2026.

More turns do not mean better answers.

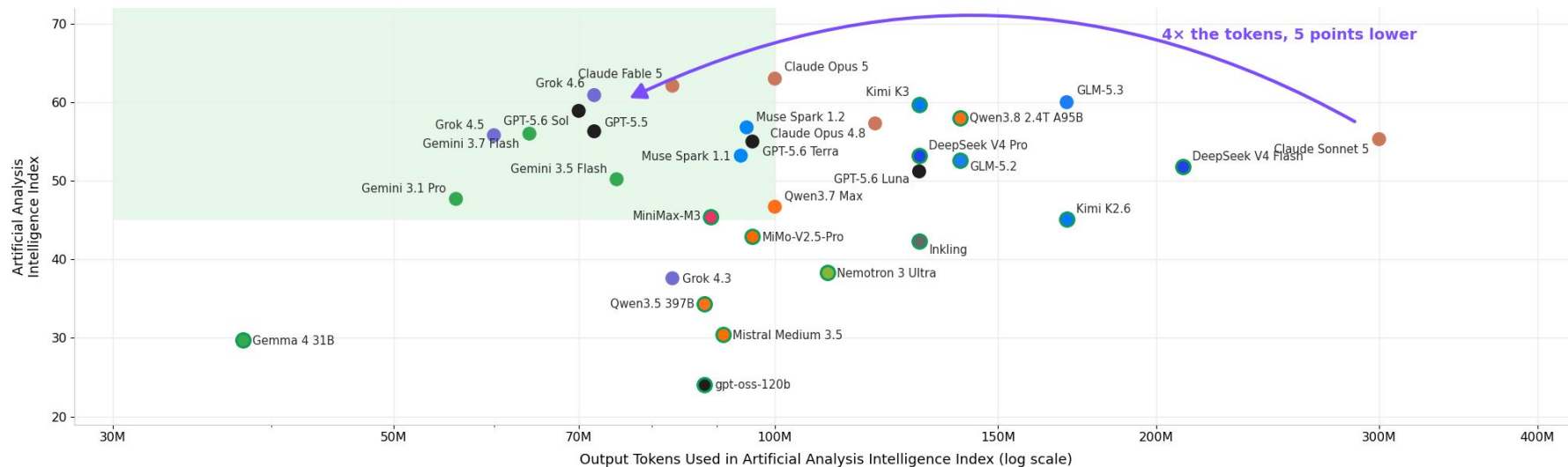
Intelligence vs. Output Tokens Used in Artificial Analysis Intelligence Index

Intelligence Index v4.1.1 · total output tokens (answer + reasoning) to run all evaluations

■ high score, low token cost

● green ring = open-weights model

Data: Artificial Analysis · retrieved 19 Aug 2026



Intelligence per Watt

The right metric for the agentic AI era

Tokens per Watt measures the currency. **Intelligence per Watt** measures the value.

$$IpW = \frac{\text{Completion} \times \text{Precision} \times \text{Efficiency}}{\text{Energy Consumed (Watts)}}$$

01

Goal Completion

Did the system fulfill the user's stated objective?

02

Precision

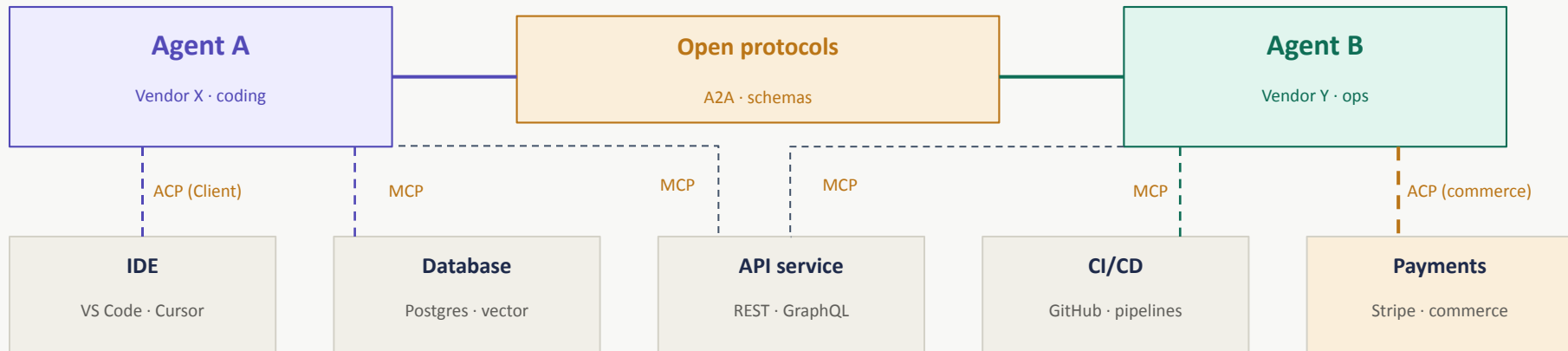
Were the outputs verifiable and free of hallucination?

03

Reasoning Efficiency

Was reasoning depth proportional to task difficulty?

Layer 4: Protocols and interfaces matter



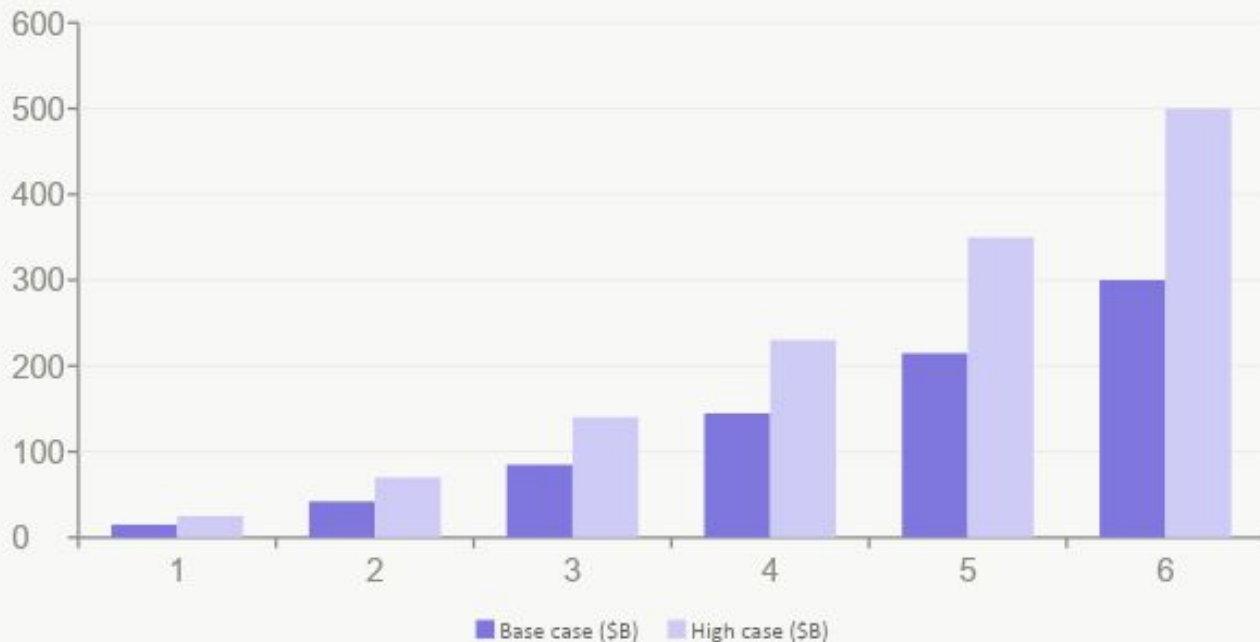
Without open standards: every connection is a custom integration.

2 agents × 5 tools × N vendors = combinatorial explosion of proprietary glue code that no one can maintain.

Open protocols collapse this to one interface per layer. That is why MCP, A2A, and ACP matter.

Layer 5: \$300–500B US agentic commerce market by 2030

Bain & Company (Dec 2025) defines agentic commerce as purchases initiated, influenced, or completed by AI agents. X-axis: 1 = 2025 → 6 = 2030.



\$300–500B

US agentic commerce by 2030

15–25%

Share of total US e-commerce

\$3–5T

Global opportunity (McKinsey)

\$67B

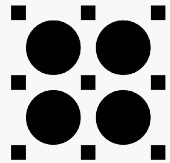
AI-influenced Cyber Week sales

Key Challenges in Agentic AI

Ten problems no single company can solve.

- | | | | |
|-----------|--|-----------|---|
| 01 | Identity & Delegated Authority
Who an agent is, who it acts for, and how authority is scoped, verified, and revoked. | 06 | Protocol Fragmentation
From no standards to too many — competing agentic and commerce protocols must be harmonized, not multiplied. |
| 02 | Reliability & Compounding Error
Per-step accuracy collapses over long task chains; failures are non-deterministic. | 07 | Economics & Efficiency
Agentic workloads burn 10–100× the tokens of chat — value per watt, not tokens per watt. |
| 03 | Security & Prompt Injection
Untrusted content hijacks agents; tool poisoning and supply-chain attacks persist. | 08 | Memory & Long-Horizon Context
Persistent state across sessions and agents, without leaking what is remembered. |
| 04 | Observability & Auditability
Tracing reasoning across hours, tools, and sub-agents to regulator-grade audit trails. | 09 | Evaluation & Benchmarks
Single-turn benchmarks miss long-horizon work; frontier agents still score below 60%. |
| 05 | Governance & Liability
Accountability for autonomous errors is unsettled; rules fragment across jurisdictions. | 10 | Multi-Agent Coordination
Delegation, negotiation, and recovery across vendor boundaries, at internet scale. |

None of these can be solved by one company — that is why a neutral foundation exists.



**Agentic AI
Foundation**

The Neutral Home Where the Open Agentic Stack Is Built

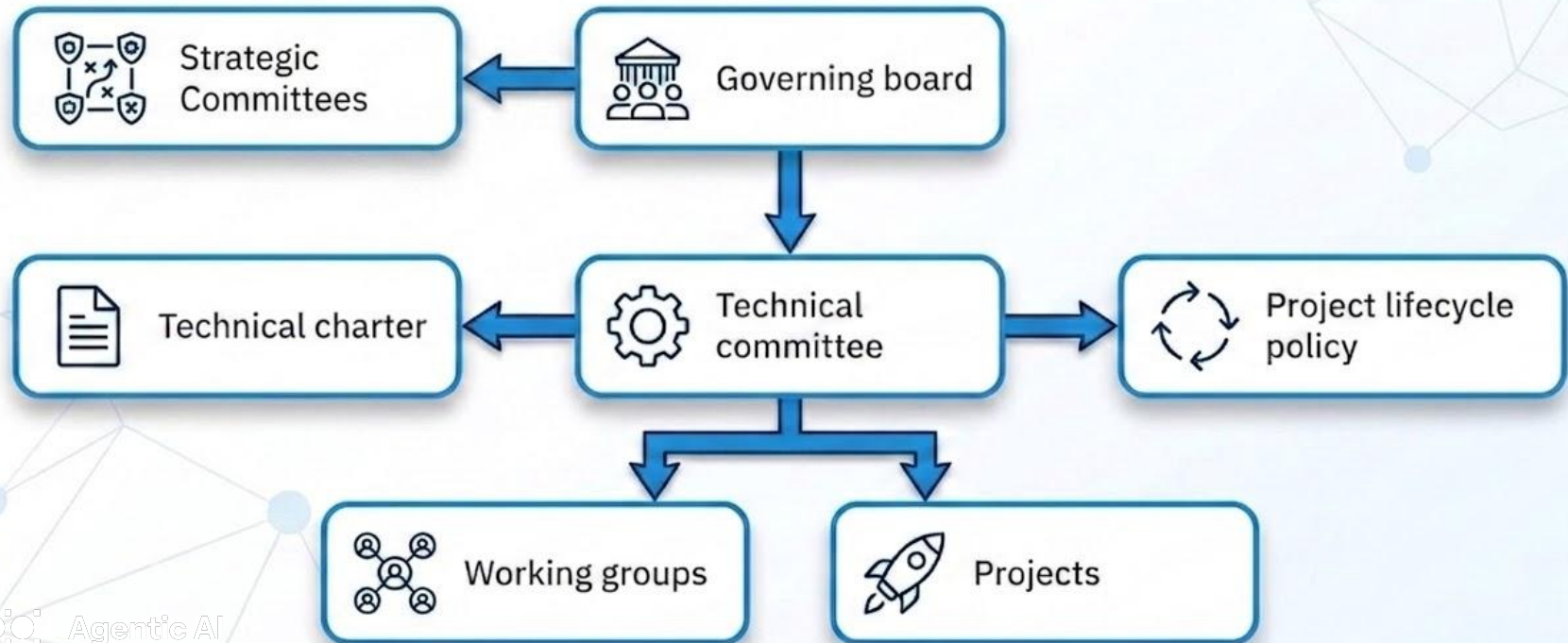
Open Protocols

Stateless Infrastructure

Neutral Governance

Governance

How we work together.



Agent = Model + Harness + Protocols + Tools

The model reasons. The harness decides what it sees, what it can do, and when it stops.

APPLICATION your product, your workflow, your domain logic

AGENTS.md · **SKILL.md** project context & portable capabilities

HARNESS *everything between the weights and the world*

Loop &
control flow

Tool
routing

Context &
compaction

Memory
& state

Sandbox &
execution

Permissions
& credentials

Observability
& replay

Errors
& retries

MCP agent ↔ tools & data

A2A agent ↔ agent

MODEL weights + tool-use post-training

STANDARDIZED TODAY

- MCP — the tool boundary
- A2A — inter-agent delegation
- AGENTS.md — project context

OPEN — WHERE THE WORK IS

- Permission & authorization manifests
- Trace and telemetry schema
- Conformance test suites
- Memory, context, tools, and models portability

COMPETITIVE BY DESIGN

- Planning, quality, & control strategy
- Compaction & retrieval policy
- Model efficiency and UX
- Pricing, scale and customers

Same weights. Different harness. Different agent. *We standardize the seams, not the internals.*

Why Companies Join AAIF

A strategic seat at the table shaping the open standards, working groups, and ecosystem that will define agentic AI.

01 Governance Influence

Nominate a representative for a shared Gold seat on the AAIF Governing Board — direct voice on strategy, projects, budget and policy.

02 Lead a Working Group

Eligible to chair or co-chair a working group — set the technical agenda on the standards enterprises will run in production.

03 Strategy Meeting

Private strategic session with AAIF leadership to shape roadmap direction and provide inputs on organization's priorities.

04 Executive Convenings

Invite-only leadership gatherings every 2–3 months, restricted to Platinum and Gold members — closed-door peer dialogue.

05 Meetups & Maintainer Access

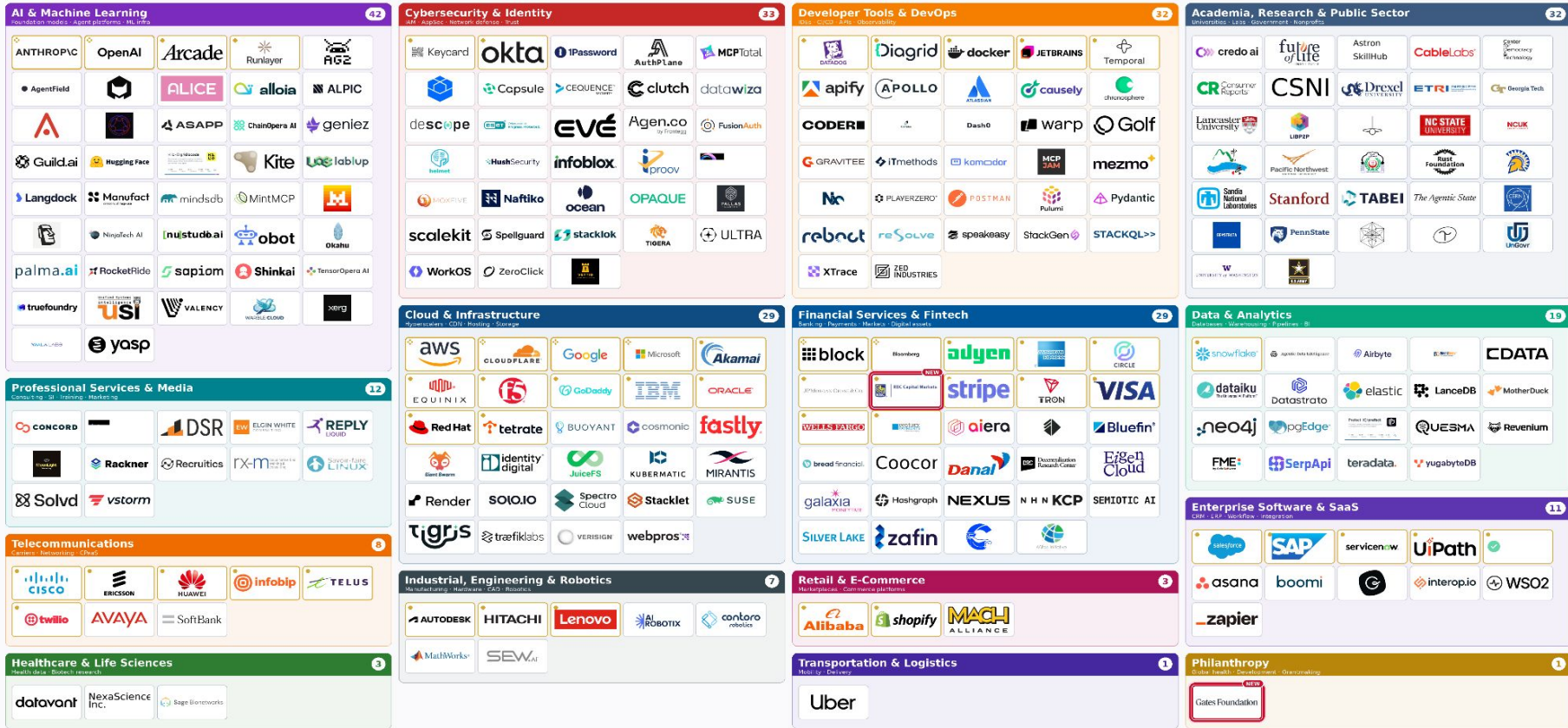
Speaker slots and venue-host opportunities at AAIF Community meetups — engage directly with project maintainers and top contributors.

06 Early Signal, Louder Voice

Early access to initiatives and ecosystem reports, and joint collaboration of marketing outreach

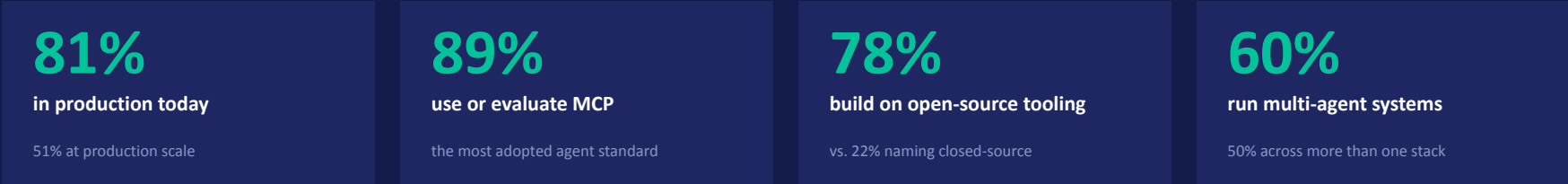
AAIF Member Landscape

270 Members across 15 industrial sectors

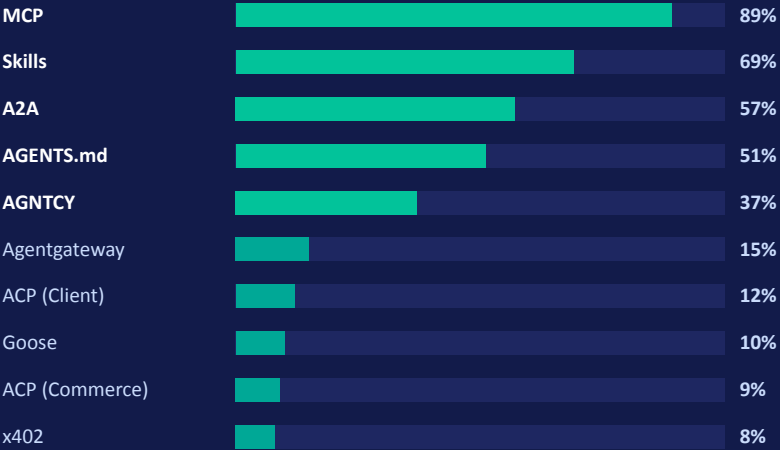


Where the agentic stack is being built.

Agentic AI Is Already a Production Market



THE OPEN STANDARDS STACK — TOP 10 ADOPTED

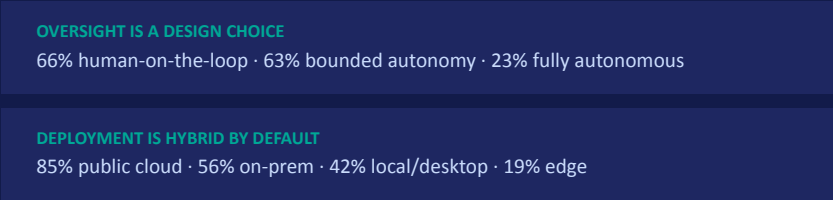
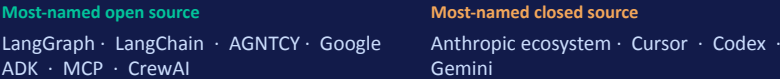


Using or actively evaluating (select all that apply, n=175). MPP follows at 6%.

OPEN SOURCE DOMINATES THE TOOLING LAYER



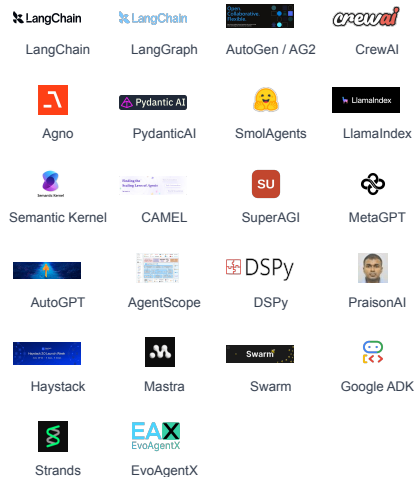
Share naming at least one tool of each type. Unaided open text (n=185), so the groups overlap.



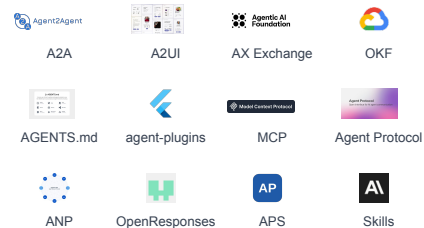
Agentic AI Project Landscape

There are 100+ open source projects we are tracking

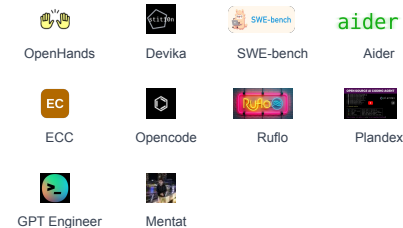
Core Frameworks & Multi-Agent Systems (22)



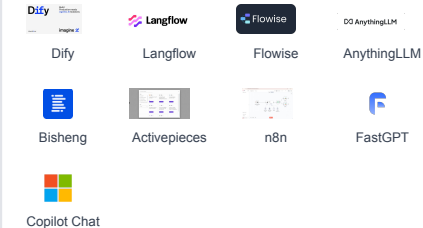
Protocols, Standards & Specifications (12)



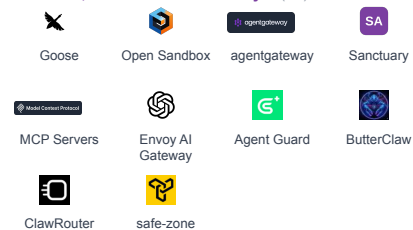
Autonomous Coding & Software Engineering (10)



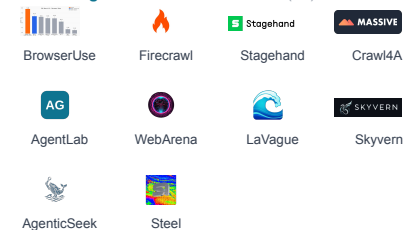
Visual Builders & Developer Tools (9)



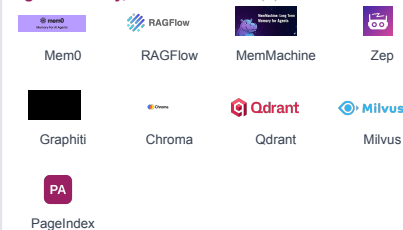
Runtimes, Sandboxes & Gateways (10)



Browser Agents & Web Automation (10)



Agent Memory, State & Context (9)



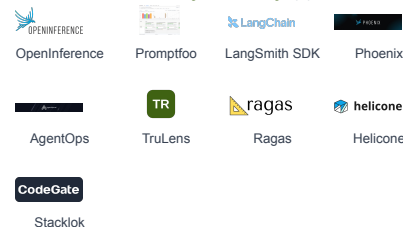
Payments, Commerce & Monetization (6)



Assistants & Community (3)



Evaluation, Observability & Security (9)



AAIF Projects

AAIF is host to top open source technology projects in agentic AI.



Model Context Protocol

Model Context Protocol (MCP) is a protocol for seamless integration between LLM applications and external data sources.



goose

An open source, extensible AI agent that goes beyond code suggestions. Install, execute, edit, and test with any LLM.



AGENTS.md

AGENTS.md is a simple, open format for guiding coding agents.



agentgateway

agentgateway enables secure, scalable connections between AI agents, models, tools, and APIs across ecosystems.

Agent2Agent (A2A)

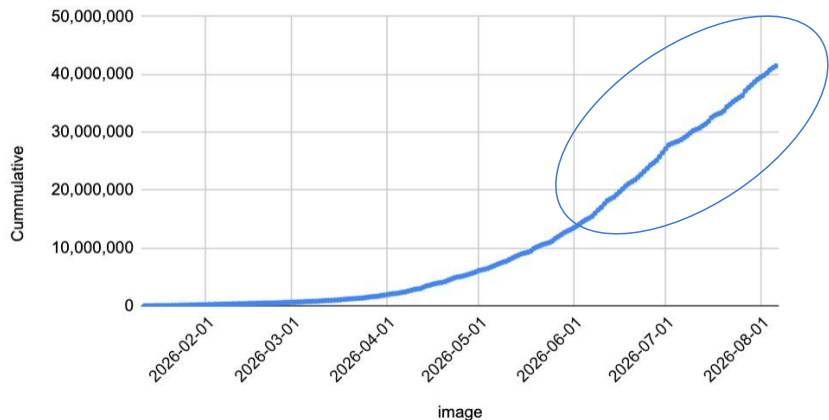
An open standard for seamless communication and collaboration between AI agents. In a world where agents are built using diverse frameworks and by different vendors, A2A provides the definitive common language for agent interoperability.

Agent Router

Agent Router is a vendor-neutral gateway for AI and agent traffic, unifying access to 16+ model providers and MCP servers with authentication, rate limiting, and observability.

What changes after a project joins AAIIF

Cumulative Image Pulls



agentgateway added x5 more image pulls since joining in June.

MCP SDK Monthly Downloads



MCP SDK Download bypassed 500 Billion/Month in August 2026.

The Seven AAILF Working Groups

Seven open working groups. Not one of these is a model problem.

01 Accuracy & Reliability

Per-step accuracy collapses over long task chains; failures are non-deterministic.

02 Agentic Commerce

ACP, UCP, x402 and AP2 must converge — or every merchant becomes a custom integration.

03 Governance, Risk & Regulatory

Accountability for autonomous errors is unsettled; rules fragment across jurisdictions.

04 Identity & Trust

Who an agent is, who it acts for, and how authority is scoped, verified, and revoked.

05 Observability & Traceability

Tracing reasoning across hours, tools, and sub-agents to regulator-grade audit trails.

06 Security & Privacy

Untrusted content hijacks agents; tool poisoning and supply-chain attacks persist.

07 Workflows & Process Integration

Orchestration, hand-off and human-in-the-loop patterns across vendor boundaries.

Member-led, open to participate — meetings, mailing lists and repos are public

Ambassador Program



140

ambassadors

6

major regions

41

countries



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[AAIF Ambassador Program](#)

The Future is Agentic. Let's Build It Together.

Five ways to start this week — scan any code.



AAIF

Who we are, our members,
and how to join.

aaif.io



Working Groups

All seven. Meetings and
repos are public.

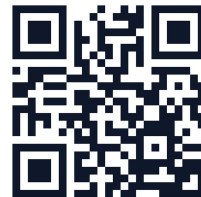
aaif.io/working-groups



Submit a Project

Bring your project to a
neutral home.

aaif.io/submit-a-project



Events

AGNTCon, MCPCon and
local meetups.

aaif.io/events



AAIF Projects

MCP · A2A · agentgateway
goose · AGENTS.md

github.com/aaif

Contribute code, technical architecture, reference implementation, designs, or industry knowledge

MCP Dev Summit



October 5-6

Toronto



November
19-20

Nairobi

AGNTCon + MCPCon



October 22-23

San Jose

Registration is Open! 25% member discount code: **COMMUNITY25**



Agentic AI
Foundation

The Future is Agentic. Let's Build It Together.